

TERRITORIAL ARRANGEMENTS OF SMALL AND MEDIUM-SIZED TOWNS FROM A FUNCTIONAL-SPATIAL PERSPECTIVE

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ABSTRACT

This contribution applies a functional-spatial perspective to the study of small and medium-sized towns. A methodology for functional analysis is developed and tested to identify towns, distinguish towns from cities, and detect town positionality within urban territorial arrangements, focusing on whether they are autonomous, networked with other towns or agglomerated with large cities. The methodology is used to compare settlement systems in two regions and two countries in Europe by looking at the existence and nature of networks between towns and cities and how towns perform in terms of population and jobs in respect of their belonging to different types of territorial arrangement.

Key words: small and medium-sized towns, large cities, urban centres, settlement systems, territorial arrangements, functional-spatial analysis

INTRODUCTION

Small and medium-sized towns (SMSTs) are important elements in urban and regional systems, local centres that provide jobs and services to the population, thus acting as vital agents in local and regional development. While many small towns are threatened by a loss of jobs, services and other centrality functions, they can benefit from interurban linkages. Therefore, it is particularly pertinent to know whether and how many SMSTs are part of urban networks as opposed to being self-standing and isolated and whether this impacts on their performance.

Despite recent inquiries shifting the focus from large cities and major urban agglomerations to include second-tier cities (Agnoletti *et al.* 2015; Camagni & Capello 2015; Cardoso

& Meijers 2016), insights into the position of SMSTs within urban systems are missing (Servillo *et al.* 2017). We attempt to bridge this knowledge gap by investigating SMSTs and their territorial arrangements. Focusing on settlement systems in their entirety, we attempt to address issues posed by Burger *et al.* (2014a, p. 1921), who urged that ‘it remains unclear (1) to what extent urban systems become more polycentric and spatially integrated and (2) whether polycentric, spatially integrated urban systems are more economically efficient than their monocentric and/or non-integrated counterparts’. In particular, we investigate which SMSTs form part of polycentric city networks, which are agglomerated within hierarchically organised metropolitan regions, and which remain self-standing and rather isolated urban centres. Furthermore, we ask how these

various locations affect their development and performance.

In this paper we first focus on the conceptualisation and identification of SMSTs and their territorial arrangements. Employing a functional-spatial perspective, we develop methods to identify SMSTs and their territorial arrangements and apply them in an analysis of settlement systems in selected countries and regions: Catalonia, Czechia, Central France and Slovenia. Finally, we provide initial insights into the performance of SMSTs in terms of population and jobs according to their membership in different types of territorial arrangements.

TOWNS' TERRITORIAL ARRANGEMENTS

Towns evolve via their relationships with other towns and cities; together, they form urban and regional systems (Berry 1964). As Rondinelli (1983, p. 385) suggests, 'the value of small urban centers is not so much in their ... sizes as in their functional characteristics'. We can productively think about towns 'in terms of influence and reach, rather than population size, density or growth' (Bell & Jayne 2009, p. 689). The functions and performance of SMSTs depend on their relations with other towns and cities (Meijers & Burger 2017). Thus towns should be studied in relation to the territorial system in which they are embedded (Andersen *et al.* 2011), focusing on 'the ways in which small cities link with other cities (and non-urban places) and the forms that these linkages take' (Bell & Jayne 2009, p. 689).

The city network theories claim that cities can benefit from synergies in complementary, reciprocal and co-operative activities and relationships (Batten 1995; Capello 2000). Consequently, polycentric urban systems have been associated with favourable conditions for economic performance. The concept suggests that some cities may perform better than others provided they can benefit from agglomeration economies developed and shared among geographically proximate cities. Meijers and Burger (2010) refer to polycentric urban regions by emphasising that external economies may not be confined

to a single urban core but rather shared among a collection of close-by and linked cities. In polycentric urban regions, physical proximity is likely to be utilised by functional linkages that reflect the 'possibilities for the 'borrowing' of certain skills and expertise from nearby urban areas' (Phelps 2013, p. 160). However, empirical studies have not yet provided a clear answer to the question of whether polycentric and networked urban regions perform better than monocentric and hierarchically organised (Burger *et al.* 2014c; Brezzi & Veneri 2015).

While most studies of polycentric urban systems and regions focus on major cities, less is known about whether and how SMSTs can benefit from networking with other towns and cities. Of particular relevance is the recently reinvented concept of 'borrowed size' (Burger *et al.* 2015; Meijers *et al.* 2016; Meijers & Burger 2017), originally outlined by Alonso (1973). Meijers and Burger (2017) neatly summarise the key premises of the concept, which refers to a situation in which smaller towns located within larger metropolitan areas perform more favourably due to the agglomeration benefits gained from their geographical proximity and functional interconnectedness with larger cities. Recent analyses of the effect of borrowed size have investigated the presence of higher-order metropolitan functions (Meijers *et al.* 2016; Meijers & Burger 2017) and high-end cultural amenities (Burger *et al.* 2015), stretching the concept of borrowing to include large cities (Meijers & Burger 2017). However, small towns and their functions and performance in relation to local labour markets and the provision of local services remain unexplored.

There are several types of territorial arrangement between small, medium-sized and large cities. First, there are smaller towns in metropolitan areas which may gain agglomeration benefits from large cities while avoiding their costs. However, fierce competition may also mean they find themselves in an 'agglomeration shadow' (Burger *et al.* 2015), which may lead to a decline in centrality functions and consequent transformation to predominantly residential areas dependent on the large city. Burger *et al.*

(2015) and Meijers and Burger (2017) found that in terms of metropolitan functions and high-end cultural amenities, larger cities are better able to borrow size than smaller cities. However, these analyses have not explored the way in which the proximity and interaction with larger cities affects the ability of small towns to retain their local functions and perform better than isolated towns.

Another option is a network of small and medium-sized towns. Here the key issue is whether such a network of SMSTs can be a substitute for the agglomeration economies of large cities (Phelps *et al.* 2001; Johansson & Quigley 2004). This hypothesis was tested for US metropolitan statistical areas (MSAs) by Meijers and Burger (2010, p. 1383), who pessimistically found that 'a network of geographically proximate smaller cities cannot provide a substitute for the urbanisation externalities of a single large city'. However, it should be noted that the analysis involved only areas with a total population of over 250,000 inhabitants, leaving many SMST networks unexplored.

Finally, peripheral, less urbanised regions, distant from major metropolises are often characterised by the presence of autonomous towns. Although towns in these regions are usually smaller in terms of populations and jobs, they often play a significant role as key nodes within extensive rural and peripheral territories. These towns cannot directly borrow benefits from nearby cities or towns; however, they are also not threatened by the competition, and they retain a somewhat provincial yet stable role as local urban centres.

Therefore, we suggest that it is worth distinguishing between the different territorial arrangements of small and medium-sized towns and testing their effect on the town's performance. We suggest that there are three basic types of SMST positions within territorial arrangements:

1. autonomous (isolated, self-standing) SMSTs (usually located in peripheral rural regions);
2. agglomerated SMSTs that are integral parts of polynucleated metropolitan areas and conurbations dominated by large cities and major metropolises; and
3. polycentric networks of several SMSTs (usually located in modestly urbanised regions outside major urban agglomerations).

A FUNCTIONAL SPATIAL PERSPECTIVE AND METHODOLOGY

Aiming to distinguish territorial arrangements of SMSTs, we first need to identify the settlements that we consider to have the status of town or city. We argue that the key characteristic of towns and cities is their role as urban centres within regional systems. This role is expressed not only through the concentration of centrality functions, such as jobs, services or amenities, but especially through their linkages with other settlements. Towns and cities that act as urban centres do so within and for urban regions. The most common form of an urban region is one that has a single node – the core town or city – surrounded by a rural hinterland (Paar 2014). In large urbanised areas, where urban nodes form strongly interconnected networks, the single-node urban regions can develop into polycentric metropolitan areas (Soja 2000). Andersen *et al.* (2011, p. 595) thus note that traditional urban analyses that concentrate on individual urban places and on an urban/rural dichotomy encounter serious limitations when used to investigate highly urbanised territories such as Denmark, where 'the functional areas of many large cities have merged to form continuous urban landscapes'. However, analyses of the Belgian urban system (Hanssens *et al.* 2014) and Randstad (van Oort *et al.* 2010) have shown that neither the Christallerian hierarchy nor the concept of an urban network (Capello 2000) or a polycentric urban region (Kloosterman & Musterd 2001) are capable of fully capturing the nature of an urban system that combines features of both an urban hierarchy and a polycentric network. Hence, as Paar (2014, p. 1929) suggests, polycentric urban regions (PUR) 'might be better viewed as a series of city-regions, each based on a major centre of the supposed PUR' (see also Burger *et al.* 2014a).

Despite these findings calling for a more nuanced view of urban regions, they do not challenge the primary emphasis in current urban study debates on the role of urban regions as the most essential functional territorial unit of urban and regional systems (Scott 2001; OECD 2002; Antikainen 2005). While analyses of urban regions have up to now predominantly focused on large cities, metropolitan areas, conurbations or megacity regions (Hall 2009; Healey 2009; Lang & Knox 2009; Turok 2009), we have argued elsewhere that such an approach omits the very existence of urban regions of small and medium-sized towns (Sýkora & Mulíček 2009), which act as the micro-level territorial systems within which the daily activities of the population occur (Hägerstrand 1987). Summing up, from the functional-spatial perspective cities and towns qualitatively differ from settlements that do not possess central functions, do not act as centres and do not form their own urban regions.

Second, urban centres are highly differentiated according to the strength and significance of their centrality functions and hence their territorial influence. We can distinguish between large metropolises, medium cities and small towns according to the particular degree of centrality that ranks or positions these cities and towns within urban hierarchies (Christaller 1933; Lösch 1941; Berry 1973; Haggett *et al.* 1977). Therefore, we employ functional-spatial perspective to differentiate between lower rank urban centres (which we interpret as being small and medium-sized towns) and upper rank urban centres (interpreted as being large cities).

Finally, we aim to classify the positionality of urban centres within settlement systems and identify their territorial arrangements: autonomous towns, towns agglomerated with large cities and towns networked with other small and medium-sized towns. We use Czechia as our example and document the individual steps in the methodology and then provide the empirical outcomes as illustration. Figure 1 presents an overview of the essential analytical steps in the functional analysis and these are discussed in detail in the next two sections.

In the analysis, we use two types of data: information on the size of each settlement and data about relations between settlements. In relation to the size of a settlement, we use the number of jobs and population size, as these data are readily available for international comparisons. We also assume that jobs are essential for the development of other functions; thus, although jobs and population do not provide the complete picture, they can present an initial perspective and broad view of settlement systems.

As the number of jobs and size of resident population do not provide a sufficient picture of a settlement's functionality in the urban and regional systems, we also analyse functional relations between settlements. These relations could involve transportation and telecommunication, commuting to work and schools, shopping and leisure trips, flows of information and goods, etc. For the purposes of this research, we chose commuting to work. We are aware that by taking a single type of relation, we abstract from the multiplicity of functional linkages (Burger *et al.* 2014b). However, the focus is on small and medium-sized cities and their territorial constellations, and commuting-to-work represents the most frequent, intensive and important relation in the daily life of the population and the operation of local micro-regions. Furthermore, we assume that other types of trips and relations are related to job commuting. Finally, the data on commuting are usually available in the required spatial detail for national and regional territories. Therefore, we use commuting flows from places of residence to places of work in the form of a matrix of the total travel-to-work flows between all settlements within the region under study (the data should, however, also include the flows across regional borders, especially in studies at the sub-national level where the regions are not perfectly self-contained travel-to-work areas).

In the functional analysis, we use the smallest settlements and data for the established territorial-statistical units. In most countries, these units are LAU2 – usually municipalities that contain central settlements that could be a city, town or village and often also include other small settlements in their

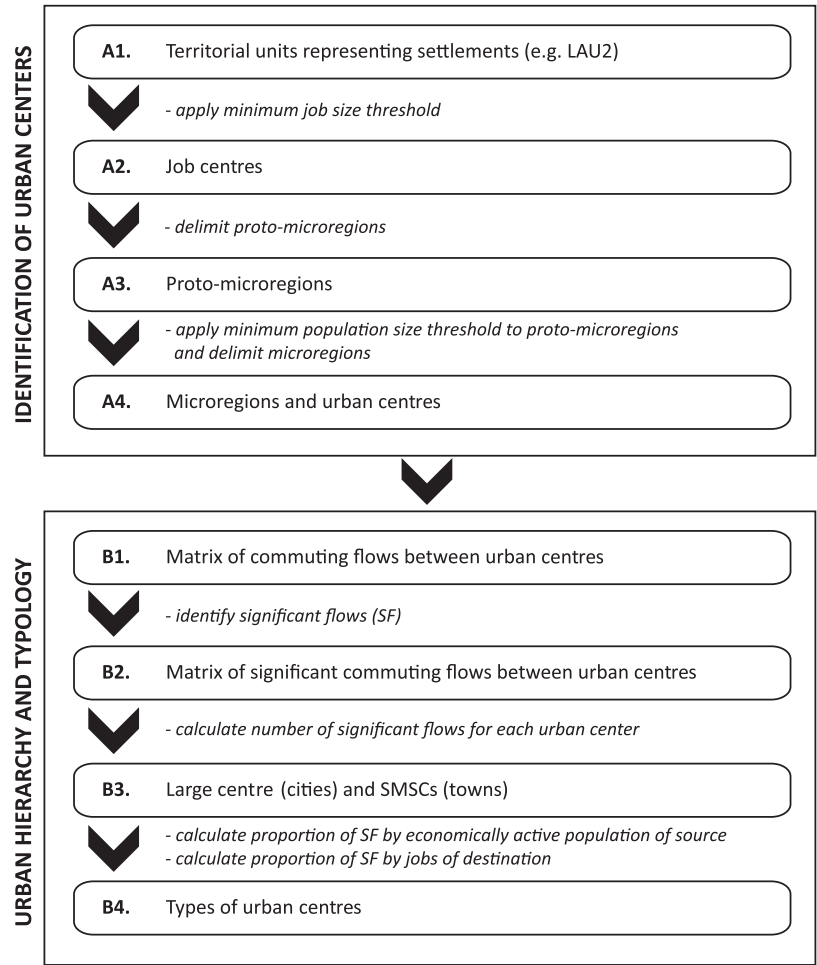


Figure 1. Functional analysis diagram - overview of analytical steps.

immediate vicinity. In the case of urban municipalities, we assume that the vast majority of the population and centrality functions, such as jobs and services, are located within the central settlement. In countries, where the real organic urban areas consist of several municipalities, such as in France, urban LAU2 should be amalgamated.

IDENTIFICATION OF URBAN CENTRES

The first task is to distinguish between settlements that act as urban centres and those that do not. Our first assumption is that towns and cities act as urban centres if they

concentrate functions that result in the formation of an urban region. As our data-sources are limited to employment and commuting data, we determine which settlements can be considered to be job centres and which are strong enough to form their own travel-to-work region.

In the first step of the functional analysis, we identify job centres as settlements that provide enough jobs to make them an important commuting destination. First, we create a set of settlements (LAU2) with a number of jobs above a minimum threshold size set discretionarily. Second, we select settlements that are the commuting destinations of the strongest outgoing flow from at

least one other municipality. Third, we intersect the two datasets to determine the job centres. There are two methodological issues to resolve. First, what is the typical threshold number of jobs needed for a settlement to become a job centre, that is, to attract significant streams of commuters? We recommend using the frequency distribution of municipalities targeted by major commuter flows by number of jobs. In Czechia, this resulted in a threshold of 1,000 jobs being selected (this may vary by country according to the national context). Consequently, we defined a job centre as being a municipality with at least 1,000 jobs that is also the main commuter destination from at least one other municipality. In Czechia in 2001, 493 of the 6,258 Czech municipalities fulfilled the criterion of having a minimum of 1,000 jobs; at the same time, 645 municipalities were classified as destinations for maximal commuting flows from other municipalities. The intersection of these two datasets resulted in a set of 367 job centres with 1,000 or more jobs that were also the major commuter destinations for at least one other municipality. These job centres include small and medium-sized towns as well as large cities.

Our other assumption is that the settlement operates as an urban centre if it forms its own microregion. Not every job centre is strong enough to attract commuters from several surrounding settlements and to form a travel-to-work microregion. The functional analysis thus continues with the delimitation of microregions, that is, territories that are related to a particular urban centre through functional ties (in our case commuting ones). Using the matrix of all the commuting flows between settlements, we link each source settlement to just one destination settlement following the direction of the strongest commuting flow. In most cases, the destinations are the previously identified job centres. If the strongest flow from a source settlement is directed to another settlement that is not one of the identified job centres, the source settlement is linked to a job centre indirectly via an intermediate settlement (for details, see Sýkora & Mulíček 2009). Using this approach, each settlement

is assigned to a job centre and this determines the proto-microregions (PMRs).

Proto-microregions should be treated as preliminary representations of microregional patterns. Proto-microregions may consist of just a job centre and one or two other municipalities. While in some instances these relatively small PMRs may emerge around a larger job centre, usually in the context of a larger urban agglomeration facing competition from large centres, there may be cases where the job centre and its region are not strong enough to provide a sufficient supply of jobs and services within the region. Therefore, it is advisable to apply the population size threshold in order to eliminate very small PMRs and dissolve their LAU2 into neighbouring job centres. Analysing the frequency distribution of a number of PMRs in population size categories in Czechia, we found a threshold size of 6,000 inhabitants (Sýkora & Mulíček 2009). Since the territories of PMRs are usually spatially fragmented, they will be consolidated into contiguous areas. Consequently, we obtain a set of territorially coherent microregions, and each microregion is organised around a microregional centre (MRC). In Czechia, the outcome for use in consecutive parts of the functional analysis was a set of 260 microregional urban centres that represent the towns and cities.

Although the functional analysis employs the concept of urban region, the primary goal is not to identify self-contained functional regions, travel-to-work or local labour market areas (cf. Van der Laan & Schalke 2001; Coombes *et al.* 2005; Karlsson & Olsson 2006; Robson *et al.* 2006). Our key aim is to determine which settlements act as urban centres. They may do so in various regional situations ranging from highly self-contained regions to areas that are strongly interconnected through a multiplicity of ties with other urban centres and that exhibit lower levels of self-containment. This latter situation is common in exposed locations within highly urbanised territories. The fact that there are higher levels of networking and lower levels of self-containment should not inhibit us from detecting micro-territorial levels within which urban centres act as regional

nodes. Therefore, in our analysis, we abstract from the self-containment because it would be biased in favour of larger interconnected areas of major urban centres and regions with highly networked urban centres, while hiding the microregional structures of small, well-connected towns. The linkages between urban centres, the interconnectedness of their regions and high-rank regional associations are reflected in the detection of a polycentric constellation of urban nodes, which we focus on in the next section.

URBAN HIERARCHY AND TERRITORIAL ARRANGEMENTS OF TOWNS

The size, centrality and position of microregional urban centres differ within the urban system. These centres include small and medium-sized towns as well as large cities. Usually population size is used to distinguish between these categories, commonly using a threshold of 50,000 inhabitants. However, given that we are adopting a functional-spatial perspective, we are interested in the territorial influence and role of towns and cities in urban and regional systems. Large cities concentrate functions provided to the population and firms in their own microregion and also to those in other microregions.

In this step of the functional analysis, we consider and analyse relations between urban centres. Combining the significant commuting flows between microregional centres with the number of jobs, we distinguish between the lower and upper tiers of the urban hierarchy, i.e., between small and medium-sized towns and large cities. Furthermore, we assess the positionality of SMSTs within the territorial arrangements of towns and cities. We begin with the matrix of commuting flows between microregional urban centres. Inspired by Van Nuffel (2007) we select only the significant commuting flows.¹ We detected 415 significant flows between microregional urban centres in Czechia.

However, some significant flows are relatively small in terms of the number of commuters and are insignificant in terms of their impact on local labour markets. We argue that significant commuting flows have to be

important in relation to the economically active population of the source urban centre and/or in relation to the number of jobs in the destination urban centre. Hence, we determine what proportion of the significant flow of the economically active population of the source centre drains a significant proportion of the economically active population from the local labour market to another urban centre. Based on an analysis of the Czech system, we decided to eliminate flows that account for less than 5 per cent and retain those representing 5 per cent or more of economically active residents; thus we selected 205 flows that are significant for the centre of their origin.

Next, we detected flows important for the job market in the destination centre. Acknowledging that there is no clear dividing line, for Czechia, we have arbitrarily set a threshold of a one per cent share of incoming commuters flow as a proportion of total jobs in the destination centre. In other national urban systems, thresholds may be set differently. In Czechia we identified 116 flows that play a significant role in the job market in the destination centre, while the remaining 89 flows are significant only for the centre of their origin.

On the basis of the existence and character of flows between microregional centres, we distinguished several types of linkages with other urban centres. First, centres that have no significant outgoing or incoming flows are autonomous. Second, microregional urban centres that are only destinations for significant commuting flows clearly show a higher rank within the urban hierarchy. Finally, many microregional centres exhibit both outgoing as well as incoming flows. If an urban centre is linked to another centre by an outgoing flow that is significant only to its own labour market, then an agglomerative relation exists. If the commuting flow is important for both the source centre and the destination centre, then a networking relation exists. There are also instances of perfectly networked pairs of towns with mutually reciprocal significant flows that are important for both centres.

Then we distinguished between microregional urban centres that are centres of

lower rank (small and medium-sized towns) and centres with higher rank in urban hierarchies and networks (large cities). Urban centres that are destinations of flows from other urban centres can be considered to be urban centres of higher functional rank. Using the matrix of significant commuting flows between microregional centres, we identified the number of incoming flows for each urban centre. Where urban centres had more than one outgoing significant flow, we added only the respective proportion (0.5; 0.33; 0.25; 0.2) of the outgoing flows to destination centres. We found that the functional position of urban centres with over 50,000 in Czechia have a value of 2.5 or greater. Hence, we detected that urban centres with populations over 50,000 and a position in the urban system of 2.5 or higher are large cities. However, there is no straightforward relationship between city size and function in the urban system. Two urban centres with high functional positions had populations of less than 50,000 and one centre with a population of nearly 100,000 exhibited a lower number of significant incoming flows. These cases were individually assessed in relation to the spatial and historical context.² Finally, we identified 14 large cities out of 260 microregional centres in Czechia.

Combining the rank of urban centre with the existence and direction of important commuting flows and their role in relation to source and destination urban centres, we identified several types of urban centres according to their position within the urban system: (A) large urban centres (cities) (LC); (B) autonomous urban centres with no significant outgoing or incoming flows (AUTO); (Ca) agglomerated urban centres with outgoing flows that are only significant locally (a significant share of the economically active population of the source centre) and are linked to a large centre (LC) – they are agglomerated with a large centre (AGLO-LC); (Cb) agglomerated urban centres with outgoing flows that are only significant locally (a significant share of the economically active population of the source centre) and are linked to another town – they are agglomerated with small and medium-sized centres (AGLO-SMST); (Da) networked

urban centres with outgoing flows that are also significant for the destination centre (with a significant share of its jobs) and are linked to a large centre (LC) – they are networked with a large centre (NETW-LC); (Db) networked urban centres with outgoing flows that are also significant for the destination centre (with a significant share of its jobs) and are linked to this destination town – they are networked with other small and medium-sized centres as source SMSTs (NETW-SMST-S); (Dc) networked urban centres with significant incoming flow(s) from other SMSTs – they are networked with small and medium-sized centres as destination SMSCs (NETW-SMST-D).

Urban centres with two or more distinct outgoing flows to multiple destinations were sorted according to the strongest flow. For urban centres that are both the destination and source of commuting flows, we compared the strength (the sum) of the incoming versus outgoing flows to classify the urban centre.

By plotting the results of the functional analysis of Czechia on a map (Figure 2), we can see the functional types of urban centres (large cities and autonomous, agglomerated and networked small and medium-sized towns), their significant relations and type of territorial arrangement. First, there are significant urban systems organised around large centres (cities) in which the SMSCs are usually agglomerated with the large city. These systems are characterised by a hierarchical dominance of large cities developed around five out of 14 large Czech centres (Prague, Brno, Pilsen, České Budějovice and Hradec Králové). A second type is an urban system organised around a large city with networked small and medium-sized centres nearby, that is, commuting flows are significant not only for the source towns but also for the large centre, which usually contains agglomerated towns (Jihlava, Karlovy Vary, Liberec, Mladá Boleslav, Olomouc, Ostrava, Pardubice and Zlín). The third type is an autonomous large city (Ústí nad Labem). The other types include urban systems consisting solely of small and medium-sized centres (towns). The most typical arrangement is that of two towns followed by a triad

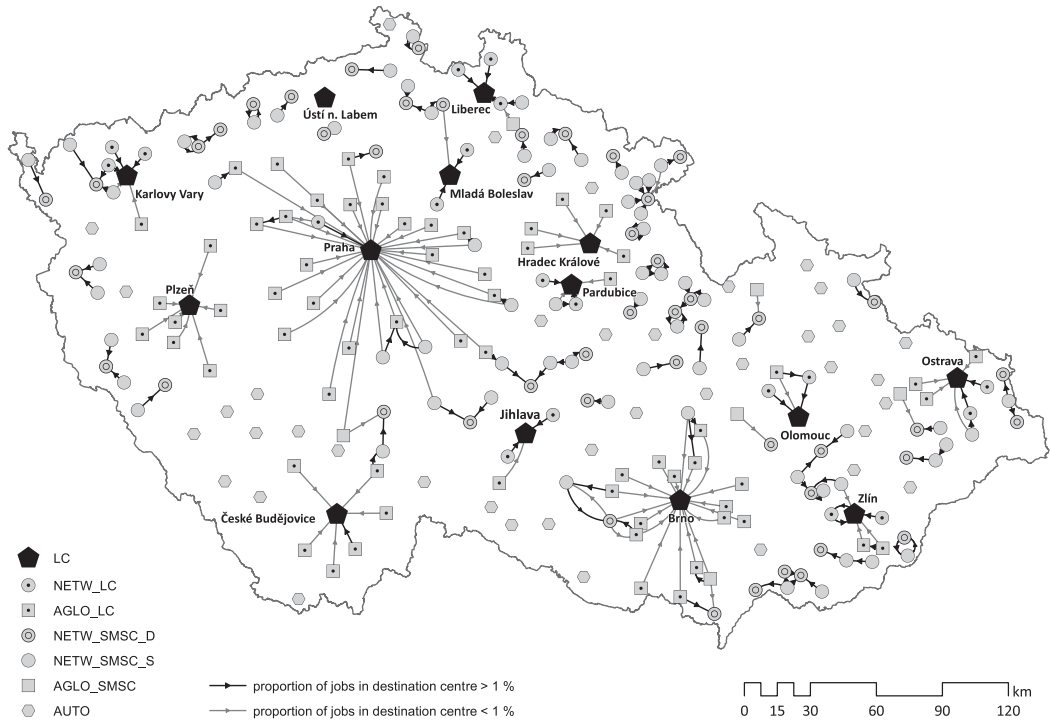


Figure 2. Types of urban centre according to positionality within territorial arrangement (autonomous, networked, and agglomerated).

consisting of a destination urban centre linked to two other towns. There are also more complex constellations of several small and medium-sized towns, where most have the status of a destination urban centre. These polycentric complexes of SMSCs can be found in eastern Bohemia and central Moravia. Finally, autonomous SMSCs are most common in peripheral rural areas, particularly in the southern part of the country and in the highland region of Jeseníky in the country's north east.

URBAN CENTRES: TERRITORIAL ARRANGEMENT AND PERFORMANCE

This section presents the results for four of the ten regions and countries (Catalonia, Czechia, Central France, and Slovenia) analysed within the framework of the ESPON TOWN project (Servillo 2014; Sýkora & Muříček 2014).³ The basic territorial unit used in the functional analyses was the

municipality (LAU 2), with the exception of France, where *unités urbaines* (aggregations of municipalities) were used instead. These units differ substantially between countries in terms of their spatial extent and population size. While the average population size of the units in Central France and Slovenia exceeds 10,000 inhabitants; in Catalonia it is 7,000 and in Czechia below 2,000. A threshold of 1,000 jobs was used to identify job centres in all the regions with the exception of Slovenia where all the significant (not only maximal) commuting flows were used to identify job centres. The population threshold value for the delimitation of microregions varied from 5,000 in Catalonia to 10,000 in Central France and Slovenia. In the case of the Central France region, the complementary threshold of minimal job size (4,000) was added. Table 1 shows the resulting number and average population size of the microregions and their centres. There is a major distinction between smaller travel-to-work areas in the former socialist countries and the

Table 1. *Microregional centres (MRC).*

Region	total population	Number of municipalities (LAU2) or alternative units	number of microregions (MR)/ microregional centres (MRC)	MRC % from all units in region	average population size of MR	average population size of MR centre total/for SMSC without LC
Catalonia, Spain (ES51)	6,343,110	946	66	6.98%	96,107 ^a	48,827/22,878
Central France region (FR24)	1,565,920	116	19	16.38%	104,395	62,673/24,292
Czech Republic (CZ0)	10,230,060	6,258	260	4.15%	39,346 ^b	24,193/14,030
Slovenia (SI0)	1,994,026	192	51	26.56%	39,099	26,348/18,265

Notes: *Unités urbaines* were the analytical units used for France. ^a46,192 without Barcelona. ^b 33,961 without Prague.

Table 2. *Number and percentage of large cities and towns according to territorial arrangement.*

Region	LC	NETW- LC	AGLO- LC	NETW- SMSC-D	NETW- SMSC-S	AGLO- SMSC	AUTO
Catalonia, Spain (ES51)	4/6%	10/15%	19/29%	10/15%	14/21%	0/0%	9/14%
Central France (FR24)	5/26%	3/16%	8/42%	1/5%	2/11%	0/0%	0/0%
Czech Republic (CZ0)	14/5%	19/7%	66/26%	49/19%	67/26%	6/2%	39/15%
Slovenia (SI0)	4/8%	10/20%	9/17%	11/22%	16/31%	1/2%	0/0%

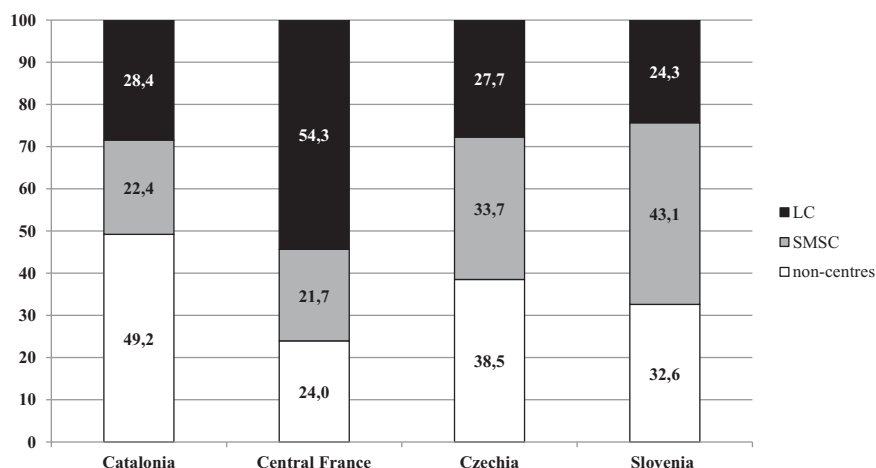
Table 3. *Proportion of the population in large cities, towns in particular types of territorial arrangement and non-centres (% of the total population of the region).*

Region	LC	NETW- LC	AGLO- LC	NETW- SMSC-D	NETW- SMSC-S	AGLO- SMSC	AUTO	Non- centres
Catalonia, Spain (ES51)	28.44	6.03	10.31	2.71	2.04	0.00	1.28	49.20
Central France (FR24)	54.33	6.48	9.26	4.09	1.89	0.00	0.00	23.96
Czech Republic (CZ0)	27.75	4.16	6.84	10.89	6.74	0.28	4.83	38.51
Slovenia (SI0)	24.34	9.89	6.33	14.76	11.93	0.14	0.00	32.61

larger regions in Central France and Catalonia with a larger commuting distance.

Having obtained a set of microregional centres, functional analysis was then used to distinguish small and medium-sized towns from large cities and identify the territorial arrangements of towns, that is, whether they are autonomous, networked or agglomerated (see Tables 2 and 3 and Figures 3–6 for an overview). The analysis of the territorial arrangements revealed substantial differences

between urban systems. In Central France, half of the population is concentrated in large centres; small and medium-sized centres account for only 22 per cent of the total, while the rest live outside the urban centres. In Czechia and Slovenia the population is evenly distributed between large centres (cities), small and medium-sized centres (towns) and non-centres. In Catalonia, nearly half the population lives outside the urban centres, while the remainder of

Figure 3. *Proportion of the population in microregional urban centres (large cities and small and medium-sized towns) and non-centres (%).*

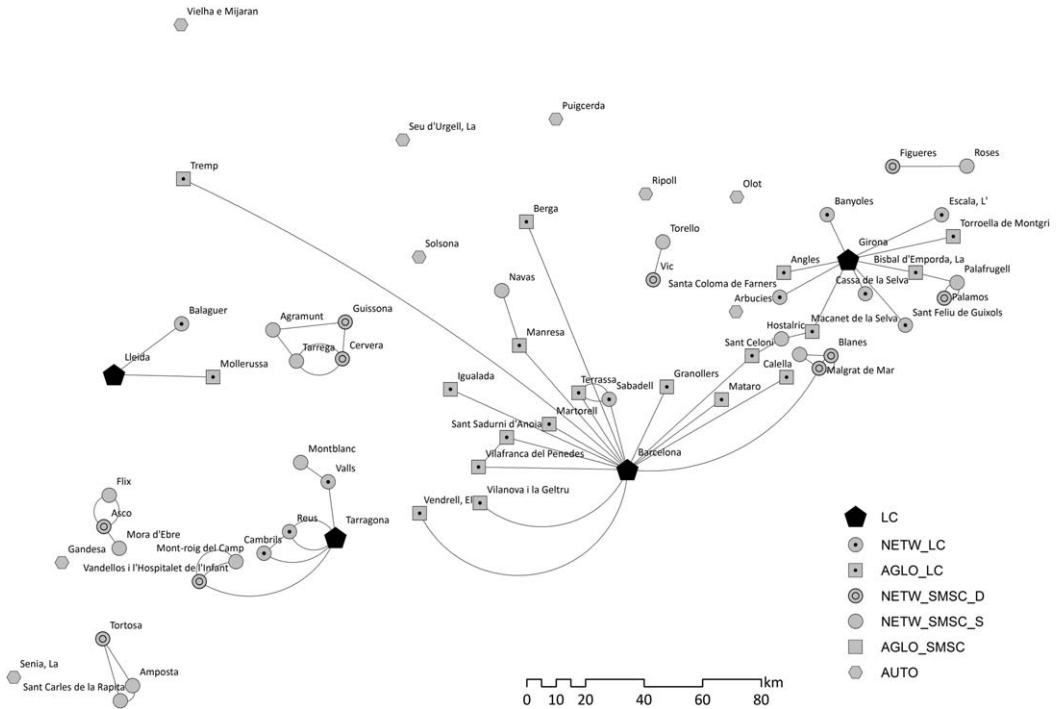


Figure 4. Catalonia: overview of territorial arrangements of type of urban centre.

the population is skewed toward large centres.

Not surprisingly, the urban systems are dominated by large centres (cities). In addition to having a large share of the total population, their key role in the settlement systems is documented by the fact that small and medium-sized centres (SMSCs) agglomerated with large centres (AGLO-LC) or SMSCs networked to large centres (NETW-LC) are the most frequent type of a town's territorial arrangement. Towns agglomerated with large cities (AGLO-LC) are typical in Catalonia (Spain) and Central France. The large number and share of SMSCs in Slovenia reveals strong functional ties with large centres; however, their relations are more reciprocal and less hierarchical, accounting for the networks with large centres (NETW-LC type). Towns are also frequently agglomerated to large cities in Czechia. However, the Czech urban system is mainly based on dense networks of towns, which is demonstrated by the dominance of networked territorial arrangements between SMSCs.

Networked towns are also significant in Catalonia and Slovenia. Autonomous towns are common in Czechia and Catalonia. There are no autonomous SMSCs in Central France due to the highly urbanised landscape. Towns of this type were not identified in Slovenia either, which can be explained by the polycentric pattern at the national level with the agglomeration of SMSCs (towns) to large centres (cities) and the polycentric arrangement of towns at the local level.

As we anticipated that the position of the SMSTs within the urban system may impact on their performance, we analysed the population and jobs growth or decline between 2001 and 2011 (2009 in France, data for Catalonia were not available). Figure 7 compares the respective categories of the functional position of towns in Central France, Czechia, and Slovenia. It clearly indicates the exclusive position and performance of large centres when compared with small and medium-sized centres. The most pronounced difference was observed in Czechia, where the number of jobs in Czech small and medium-sized centres

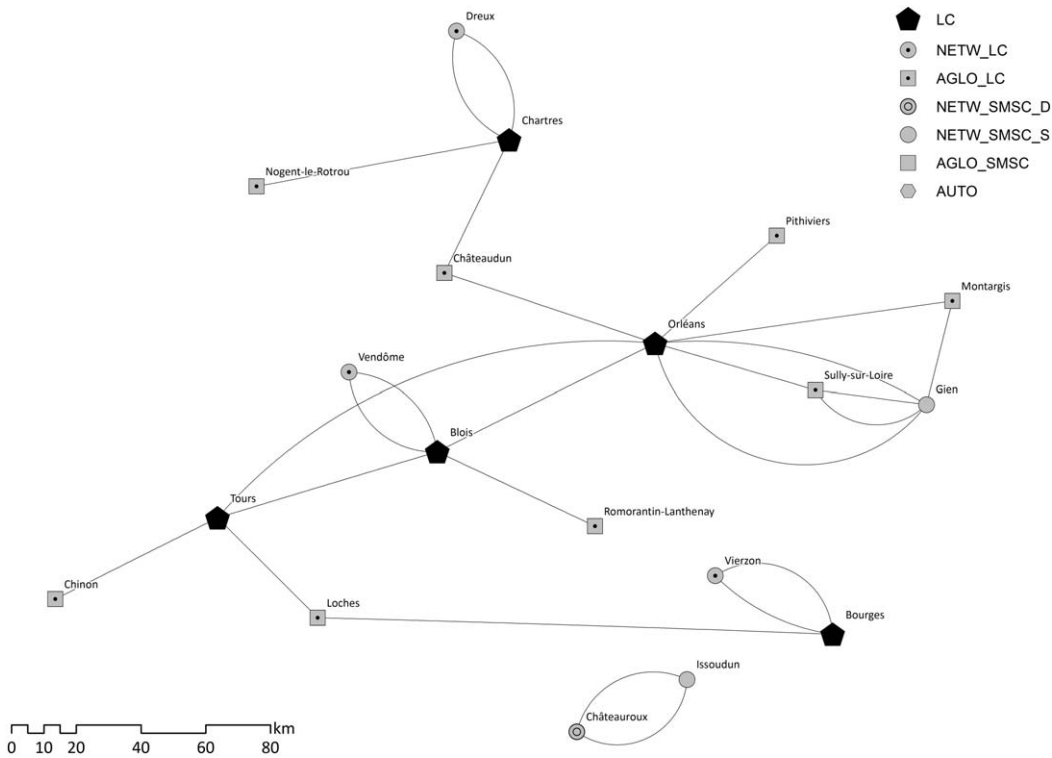


Figure 5. Central France: overview of territorial arrangement of type of urban centre.

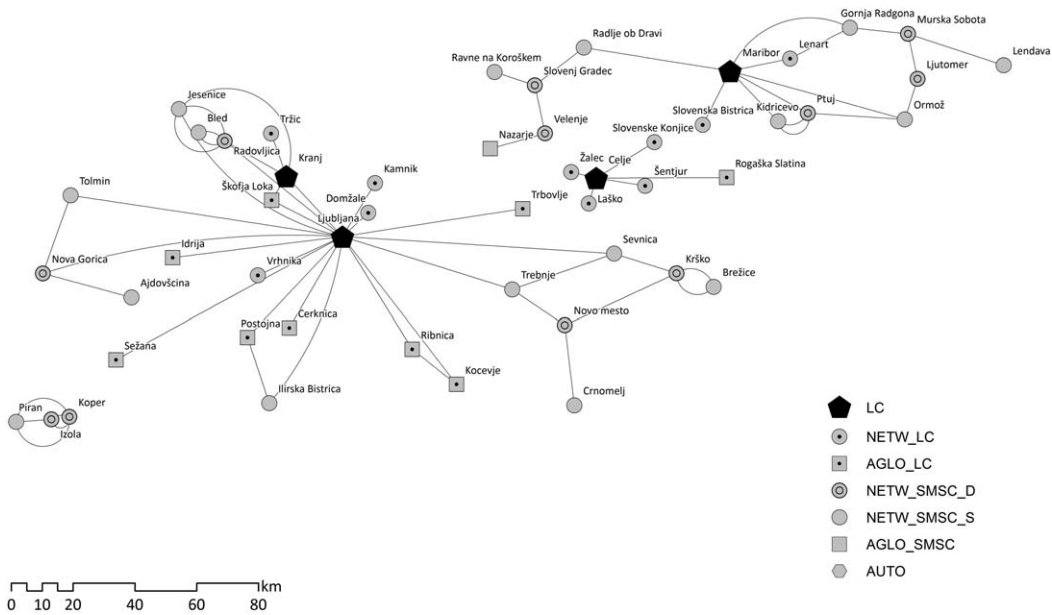


Figure 6. Slovenia: overview of territorial arrangement of type of urban centre.

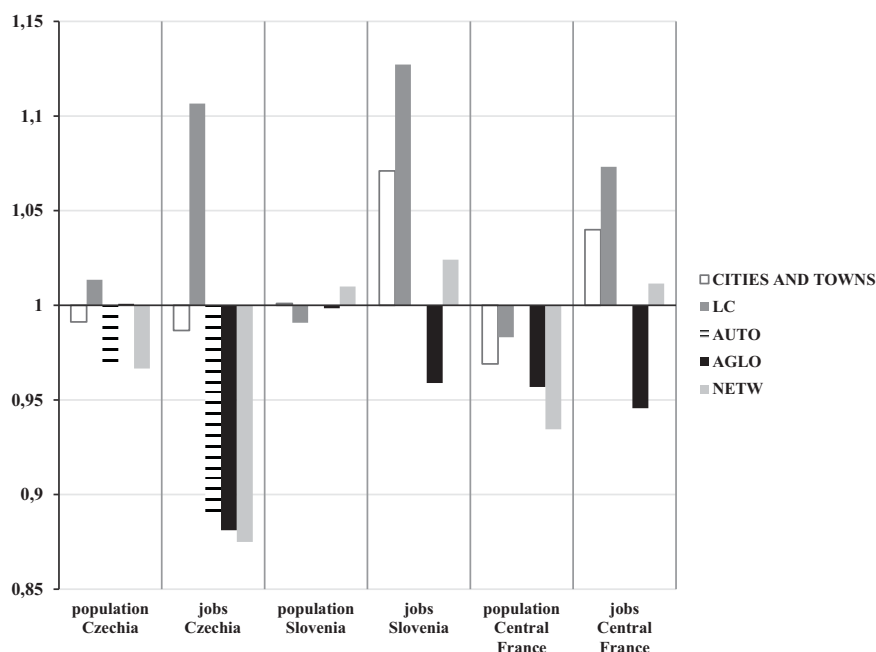


Figure 7. Population and job performance in Central France, Czechia and Slovenia.

declined by 12 per cent, while jobs in large centres increased by 11 per cent. A similar change was observed in Slovenia with 13 per cent job growth in large centres, 2 per cent job growth in networked towns, and a 4 per cent job decline in agglomerated towns, which can be explained by the competition from large cities and transformation of agglomerated towns from places of production and services into more residential suburban nodes. A similar development was recorded in Central France, with 7 per cent job growth in large centres, 1 per cent job growth in networked towns and a 5 per cent decline in jobs in agglomerated towns. While the changes in job distribution were quite significant, the distribution of the resident population reminded stable. A minor trend, however, could be observed in the stronger population decline among networked towns compared to agglomerated towns in Czechia and Central France.

As the growth trends have not shown any major differences between the territorial arrangements of different types of towns, we have also individually assessed subgroups of SMSCs agglomerated to or networked with large centres and/or small and medium-sized

centres. However, there were no significant differences between these more detailed functional categories of SMSCs. There is a great variability in the population and job performance of individual urban centres (Sýkora & Mulíček 2014) suggesting effects of plurality of diverse contextual factors. A multivariate analysis could identify the key structural conditions affecting the performance of towns and cities, of which their position within urban systems is just one. However, this task is beyond the scope of this paper. Furthermore, the conditionality of performance is so complex that the quantitative view shall be supplemented by more qualitative case studies of individual towns.

CONCLUSIONS

In the introduction to this paper, we argued that the small and medium-sized towns and their spatial constellations should be an integral part of our understanding of urban and regional systems. While they account for a substantial proportion of the population (20–45% in the regions investigated), they

are largely omitted from analyses. Using a functional-spatial perspective that emphasises the relations between settlements and cities, we have shown that many relatively small towns play an important role as local urban centres by providing jobs. Consequently, they attract substantial commuting flows from smaller village settlements in their hinterland, thus creating local microregions that are important geographical units in the population's daily life.

Referring to recent discussions on urban networks and polycentric urban regions, we stated that very little is known about the position of SMSTs within urban systems. In this respect, we sought to detect SMSTs which are isolated and autonomous, those SMSTs that are networked with other cities and towns, and analyse whether that is in hierarchically organised systems dominated by major cities or more polycentric networks, some of which may be solely based on SMSTs. The key contribution of our paper is the methodology for the typology of towns according to their positionality within urban systems and distinct territorial arrangements by focusing on whether they are autonomous, networked with other towns or agglomerated with large cities. The methodology provides a tool that can be used not only to learn about the structure and composition of various settlement systems, but also lays the ground for a comparative analysis of how towns perform according to their membership in different territorial constellations of towns and cities.

We used this methodology to empirically analyse and briefly compare the territorial arrangements of urban centres (towns and cities) in four European regions and countries. Furthermore, we conducted a simple analysis of population and job performance to indicate the effects of autonomous, agglomerated and networked positionalities within urban systems. The comparative analyses of settlement systems in Catalonia, Czechia, Central France and Slovenia indicate variability between the countries and regions in terms of the size of the urban centres and their urban regions as well as the various territorial arrangements existing within their urban systems. For instance, while networks

of small and medium-sized centres account for 27 per cent of the country's population in Slovenia and 18 per cent in Czechia, they contain only 6 per cent of the population in Central France and 5 per cent in Catalonia, where nearly half the population lives in non-urban settlements and 45 per cent in large cities or SMSTs related to large cities.

The population and job performance analysis shows that in the three regions and countries investigated (Czechia, Central France, and Slovenia) there is a major difference between large cities and small and medium-sized towns but no major differences were revealed between SMSTs according to their functional types. However, this was just an initial observation. The scope of this paper did not allow for a more advanced and nuanced analysis of town performance in relation to positionality within the territorial arrangements and in relation to other factors, such as the effects of the regional and local economy, local culture, institutional set-up, and human and social capital.

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Notes

1. The algorithm of significant flows detection compares the distributions of actually observed flows with ideal configurations. We selected the five highest outgoing flows from each microregional centre, calculated the percentages of their sums and ranked them from the highest to lowest. Then, we correlated the real relative distribution of these flows with five ideal types of distributions:

1. significant flow: 100%, 0%, 0%, 0%, 0%
2. significant flows: 50%, 50%, 0%, 0%, 0%
3. significant flows: 33%, 33%, 33%, 0%, 0%
4. significant flows: 25%, 25%, 25%, 25%, 0%
5. significant flows: 20%, 20%, 20%, 20%, 20%.

By reflecting the highest correlation between the set of observed flows and each of the five sets of ideal flow distributions, we determined the number of significant flows.

2. In Czechia, there were two urban centres with populations of below 50,000 but with positional values of over 2.5. Náchod has a population of 20,000, and it benefits from a network of several small urban centres nearby; we did not include this town among the large centres (cities). The other centre (Mladá Boleslav) has a population of 44,000 and a similar number of jobs. Since it is a very powerful job centre comparable to cities with a population of 70 to 80,000, we included Mladá Boleslav among the large centres (cities). However, there are seven urban centres with populations of over 50,000, which all have lower positions in the urban system (0 to 1). Most of these centres are in the vicinity of larger cities, which have either been agglomerated or have little regional influence due to competition from stronger and larger cities nearby. Hence, we did not consider these centres to be large cities. One exception is Ústí nad Labem, the regional capital of Northern Bohemia, which has a population of almost 100,000 and approximately 50,000 jobs. Because this city is not the destination of any significant flows from other microregional centres, we classified it as autonomous. The city does not attract any significant labour-related commuting flows from other microregional centres because its impact field is restricted physically by mountains and a border with Germany. Apart from having poor accessibility from microregional centres nearby, these centres are medium-sized towns that have stronger autonomy over their local labour markets. Although Ústí nad Labem has a weaker direct impact on the labor markets of other centres, it has a strong local microregional labor market and, at the same time, it is a strong political centre that impacts on the broader region. Therefore, we included Ústí nad Labem in the set of large centres. Out of the 260 microregional urban centres, we identified 14 large cities (centres).
3. We acknowledge the contribution of our colleagues with the ESPON TOWN project, who assembled the data from the case study regions and conducted the initial analyses to determine the typologies of towns and cities in the

study area using our methodology. We carried out revisions, adjustments and further analyses for comparative purposes.

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